Paul Parisot

Software Engineer - C++ & Data Systems

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About

Systems and data software engineer bridging the gap between advanced technical execution and analytical business strategy. Specialized in high-performance C++ backend architectures, robust Unix/Linux environments, and data stream integrity. Strong communicator experienced in interpreting complex functional requirements and translating them into scalable technical designs.

Skills

Core Languages: C, C++ (C++14/17/20), Go, Python scripting, Advanced Bash/Shell
Data & Databases: SQL (Oracle, PL/SQL engines), Relational Database Design, High-Volume Data Streams
Unix/Linux Systems: POSIX APIs, Multi-threading, System Services, Secure Resource Administration
Engineering Tooling: Git, CMake, GCC, GDB, Valgrind, Jira, CI/CD pipelines, Test-Driven Development (TDD)

Professional Experience

Faurecia / Forvia - Software Engineer (C++ / Linux Systems)

- Designed and optimized robust backend C++ software modules processing real-time telemetry data streams.
- Maintained absolute data integrity and handled strict execution constraints within critical industrial environments.
- Translated product requirement specifications into functional code using Agile/Scrum methodologies and **Jira**.
- Ensured long-term codebase reliability by enforcing strict **TDD (Test-Driven Development)** and regression testing.

Gno.land - Core Software Engineer (Distributed Architectures)

- Refactored critical sub-systems of a distributed execution engine, maximizing concurrent transaction efficiency.
- Audited production code to resolve **6 high-impact architectural bugs** (memory leaks, thread-safety issues).
- Created automated validation pipelines to guarantee system-level protocol compliance across distinct networks.

Epitech / IONIS - Systems Programming Instructor & Code Auditor

- Taught advanced C/C++ architecture, pointer arithmetic, and Unix system programming to over **400 students**.
- Developed strong cross-functional communication skills by acting as a regular technical advisor and code auditor.
- Evaluated software designs based on memory management correctness, performance profiling, and error-handling paths.

Data Engineering & Systems Projects

Sledge – Cross-Cloud Data Migration & Orchestration Engine

Co-authored a backend data integration platform in Go tailored for high-volume, secure cross-network transfers.
Designed high-performance abstraction layers to process complex relational database schemas without interrupting live data feeds.
Optimized queries and pipeline throughput to guarantee strict data accuracy and zero state loss during multi-stage migrations.
Skills: Go, Data Streaming, Relational Databases, Algorithmic Integrity, Network Robustness

rodox – Low-Latency Linux Networking Engine

Built a modular, asynchronous network communication engine in C++ leveraging a cache-friendly, data-oriented design.
Bypassed runtime OS allocation bottlenecks by engineering thread-safe **custom static pool allocators**.
Hardened network IO modules against failure using **GDB** and **Valgrind (Memcheck)** to eliminate all undefined behaviors.
Skills: C++, Boost.Asio, TCP/IP, Linux APIs, Performance Profiling, Memory Safety

Education

McGill University, Montréal 2025 – 2026
Certificate in Management
Specialization: Technical Leadership & Business Strategy

Epitech, Paris 2022 – 2027
Master's Degree in Information Technology (Master's level diploma)
Level-7 Certification - Expert in Software Engineering
Specialization: Systems & Software Architecture